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Phasors w/ Second Order Filter

Description

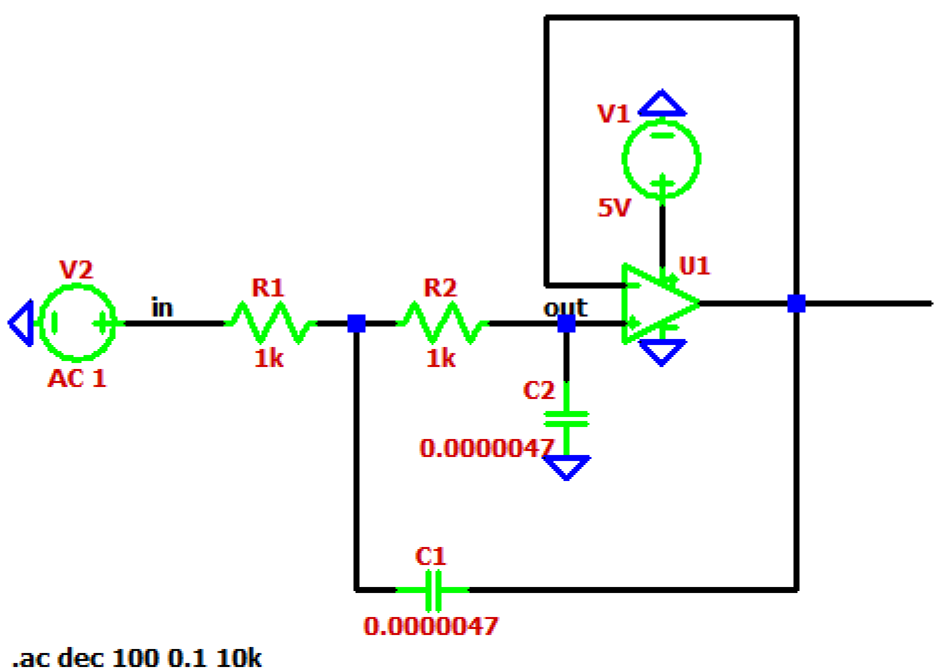
What building block are you applying this concept to?

The second-order active low-pass filter

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The output voltage phasor and the complex transfer function will be derived. By converting the circuit components into their complex impedances and applying nodal analysis, I will determine how the filter filters a specific high-frequency noise signal

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

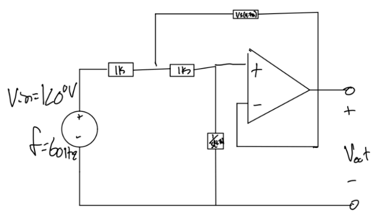
Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

H(s) = 1/(s^2R^2C^2+2SRC+1)

s=jw

w=2*pi*f

Calculate a numerical value for your circuit variable(s).



$\omega = 2\pi f = 377 \frac{rad}{s}$

$$H(s) = \frac{1}{s^2(R^2C^2) + s(C(2R)) + 1} \rightarrow s = j\omega \rightarrow H(j\omega) = \frac{1}{-\omega^2R^2C^2 + \underbrace{j\omega 2RC}_{j\omega 2RC = j3.544} + 1}$$

$$H(j\omega) = \frac{1}{-2.139 + j3.544} \rightarrow \|Denom\| = \sqrt{2.139^2 + 3.544^2} = 4.139$$

Second Quadrant

$$\theta = 180 + \tan^{-1}\left(\frac{3.544}{-2.139}\right) = 121.12°$$

$$H(j\omega) = \frac{1\angle 0°}{4.139\angle 121.12°} = 0.2416\angle -121.12°$$

$$V_{out} = H(j\omega) \times V_{in} = (.2416\angle -121.12°)(1\angle 0°) = \underline{0.2416\angle -121.12° V}$$

||V(out)|| = .242V

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

We will measure the magnitude of V(out) across a range of frequencies to verify the cutoff frequency f and the filtration at 60 Hz.

Simulation

Describe your simulation in 1 to 2 sentences.

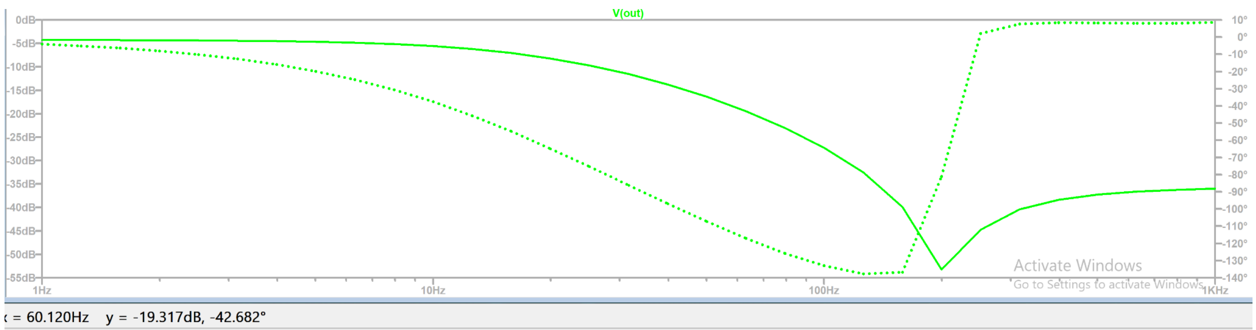
We will use an AC Sweep analysis on the second order active low-pass filter to generate a Bode plot of the circuit's frequency response. This will visually confirm the 0 dB passband for the DC temperature signals and the expected roll-off for higher-frequency AC noise.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

In the schematic I used a .tran 1m and .ac dec 1 10 1000 command. This is to do an AC sweep of the filter and find the dB in respect to Hz.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

As you can see at the bottom of the graph, at 60 Hz, it is -19.37 dB.



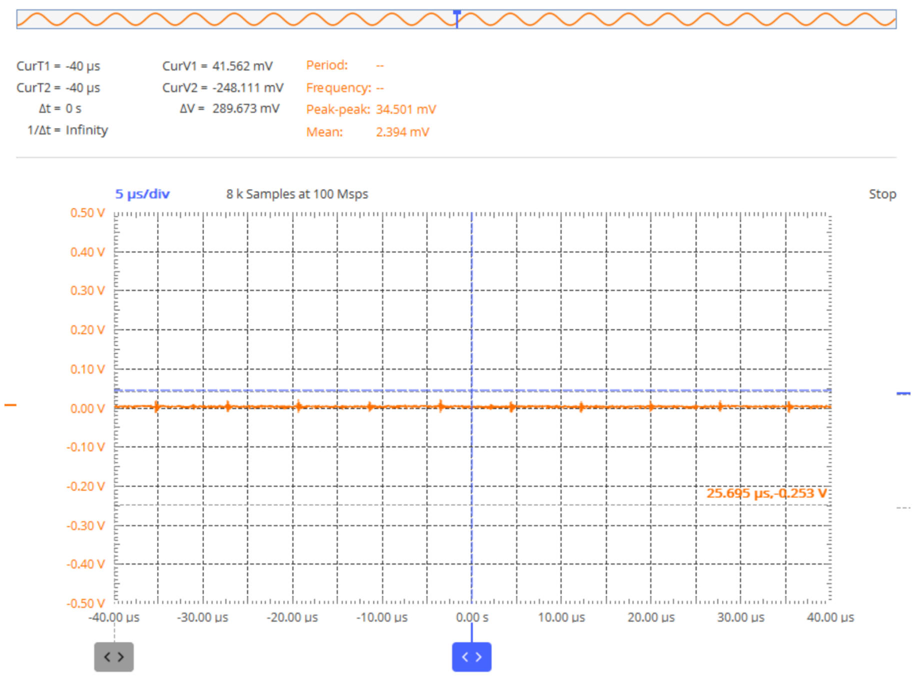
Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

We will measure the output voltage on an oscilloscope to calculate the gain. The circuit utilizes an OP484 op-amp. The experimental setup isolates the filter, temporarily disconnecting the differential amplifier from the input and the non-inverting amplifier from the output.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

As you can see at the top, the voltage across the filter is 34.5 mV which is very close to the 32mV expected.



Discussion

Discuss how this analysis helped you in your design.

Performing phasor analysis on the second-order active low-pass filter was useful to understanding the filtration of the circuit's noise. Without good filtering, the noise could cause the oscillator circuit to switch randomly, resulting in a jittery, unstable PWM signal and fan stutter. The phasor analysis confirmed mathematically that the selected R and C values provide an extremely low transmission gain about 0.032V at 60 Hz, ensuring the comparator relies strictly on the clean, filtered DC temperature signal.

The analytical output voltage magnitude at 60 Hz was calculated to be 0.242 V. The simulation produced a value of 0.107 V, resulting in a percent error of 55.8% relative to the analytical result. The experimental measurement yielded 0.0345 V, corresponding to a percent error of 85.7%. The discrepancies are attributed to non-ideal effects such as op-amp bandwidth limitations, component tolerances, and parasitic elements in the physical circuit. Despite these differences, all results confirm significant attenuation of the 60 Hz signal.

Second Order Active Filter

Description

What building block are you applying this concept to?

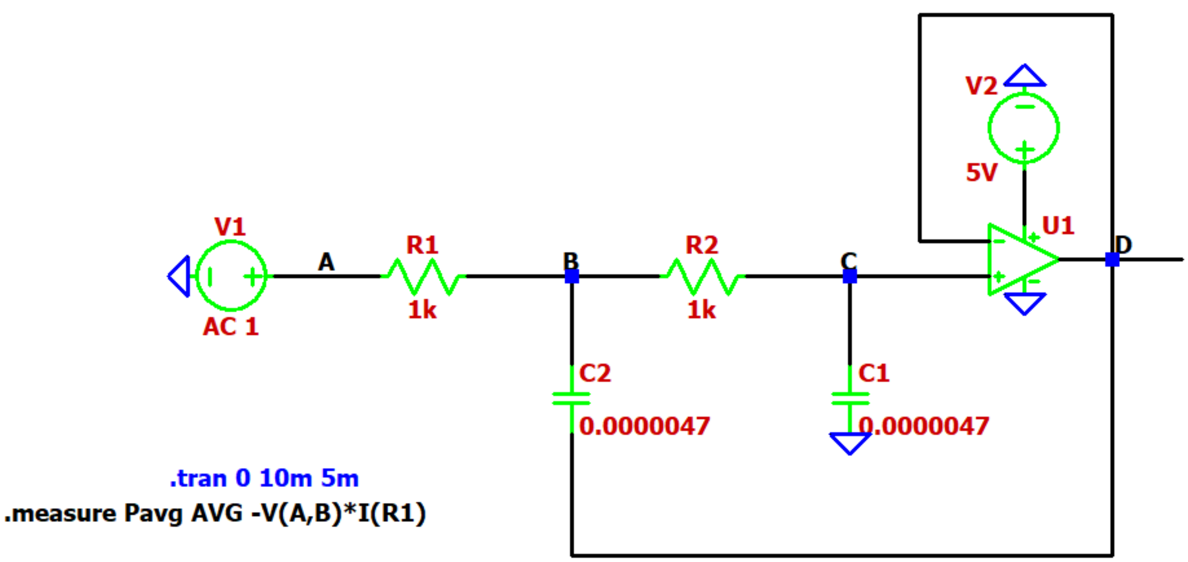
A second-order active low-pass filter is applied after the input stage to smooth the signal and remove high-frequency noise before further processing.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The cutoff frequency fc will be derived using the resistor and capacitor values in the filter. This is done by applying the standard transfer function of a second-order active filter.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.

- Input: Vin (labeled “in”)
- Output: Vout (op-amp output)
- R1=1kΩ, R2=1kΩ
- C1=4.7μF, C2=4.7 μF
- Op-amp configured as a voltage follower (unity gain)



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

For a second-order active, the transfer function is:

H(s) = 1/(s^2*R1*R2*C1*C2+s(R1C1+R2C1+R2C2)+1)

The cutoff frequency is:

Fc = 1/(2*pi*sq(R1R2C1C2))

Calculate a numerical value for your circuit variable(s).

R1=R2=1kΩ=1000Ω

C1=C2=4.7μF=4.7×10^−6F

Fc = 33.9Hz

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

To verify operation, the following will be measured:

- Output voltage magnitude Vout vs frequency
- Cutoff frequency (−3 dB point)
- Roll-off rate (expected −40 dB/decade)

Simulation

Describe your simulation in 1 to 2 sentences.

An AC sweep simulation was performed in LTspice to obtain the frequency response of the filter stage by plotting the magnitude of the output V(out) versus frequency.

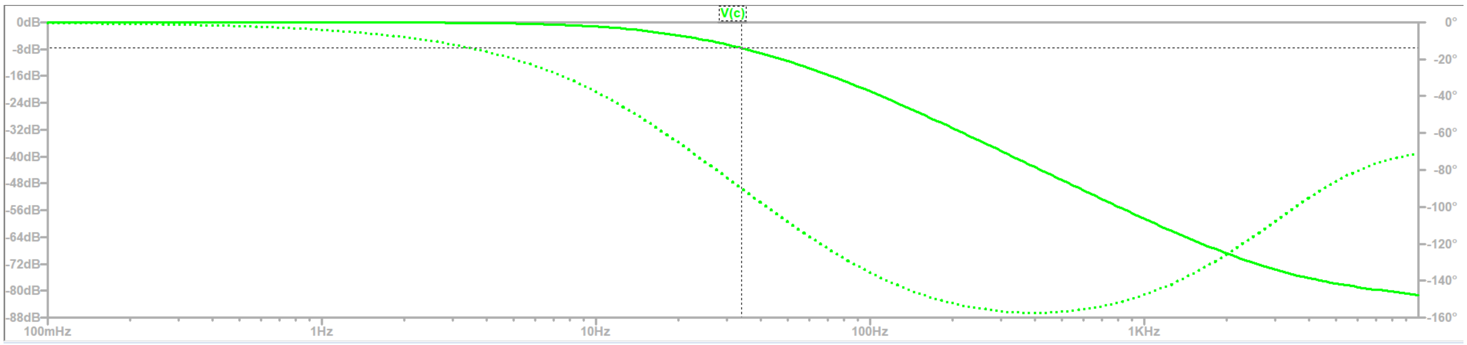
If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

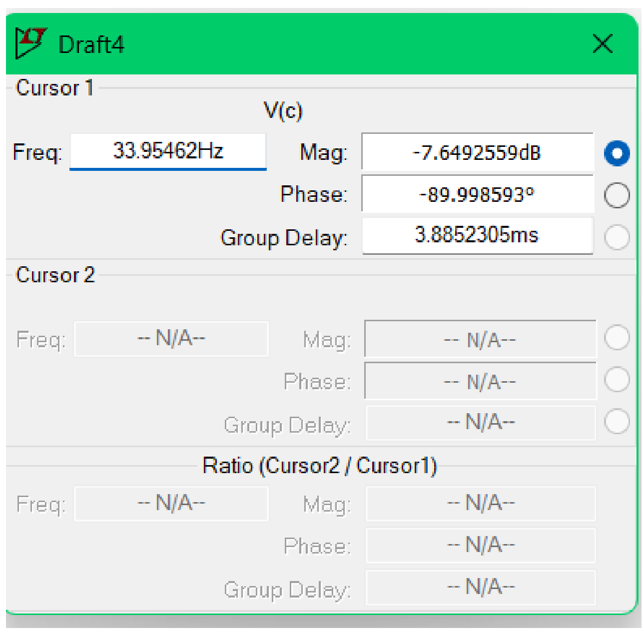
Our simulation differs slightly from the full system schematic used in MS3 because we replace the output of the differential amplifier with an AC small-signal voltage source. This modification allows us to perform an AC sweep and obtain a valid Bode plot of the second-order active filter stage without interference from nonlinear or time-dependent components later in the circuit.

Specifically, we use an AC source with magnitude 1 V and apply the simulation command .ac dec 100 0.1 10k to analyze the frequency response. The remaining stages of the circuit, such as the oscillator and transistor, are omitted in this simulation because they are nonlinear and would invalidate the assumptions required for small-signal frequency analysis.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

The resulting Bode plot shows a second-order low-pass response with a flat gain at low frequencies, a cutoff around ~30–40 Hz, and a roll-off of approximately −40 dB/decade with phase shifting toward −180°.





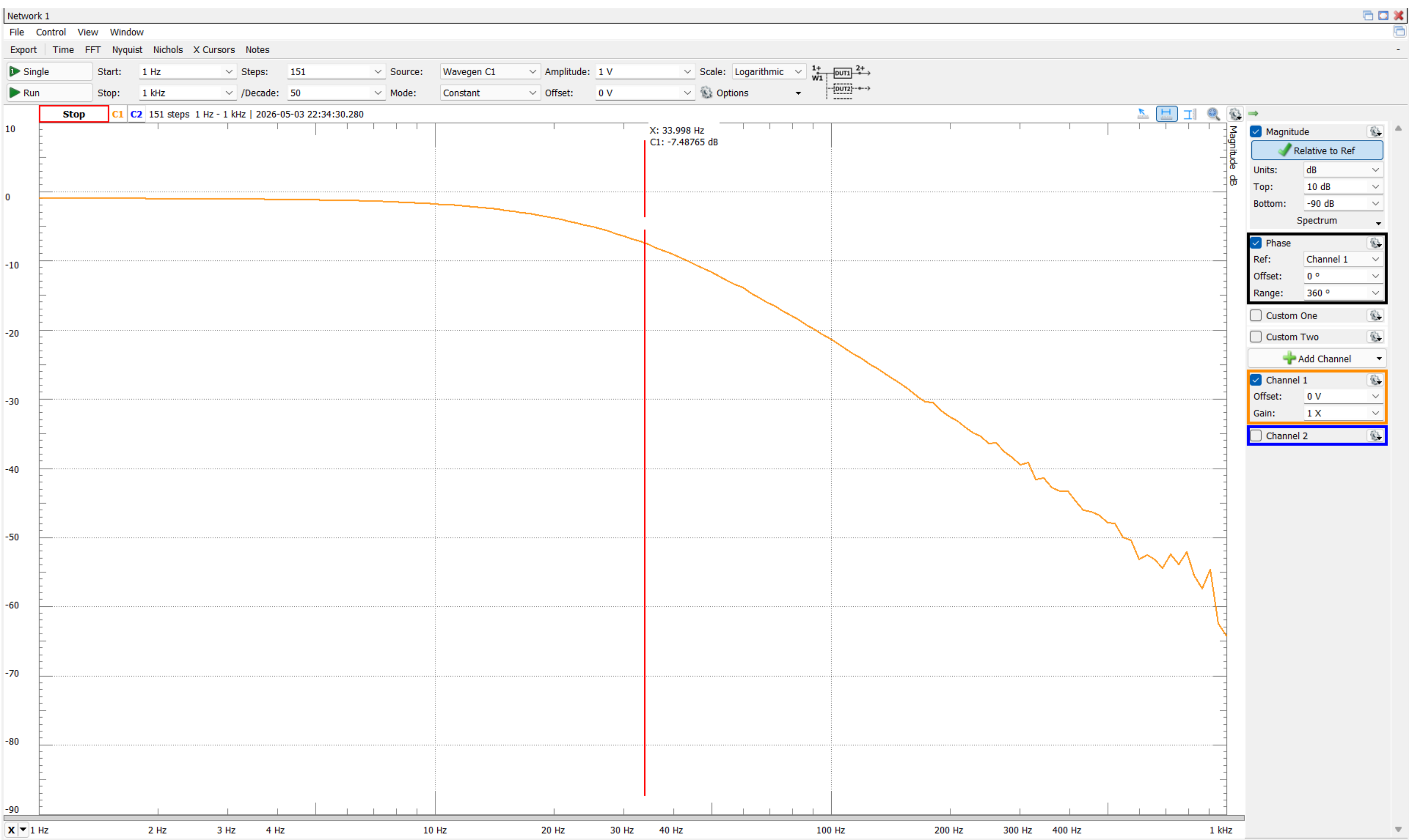
Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

The experimental measurement was conducted using the Analog Discovery Network Analyzer to perform a frequency sweep from 1 Hz to 1 kHz and obtain the Bode plot of the second-order active filter. The circuit used an OP484 operational amplifier configured as a unity-gain buffer. The measured response showed a cutoff frequency of approximately 34 Hz, along with a roll-off rate close to −40 dB/decade, consistent with second-order filter behavior.

Compared to the simulation, the experimental response exhibited slight deviations in the cutoff frequency and magnitude due to non-ideal effects such as component tolerances, op-amp bandwidth limitations, and parasitic elements introduced by the breadboard. Despite these differences, the overall shape of the response and attenuation characteristics closely matched the expected behavior.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.



Discussion

Discuss how this analysis helped you in your design.

The second-order active low-pass filter improved the overall system by providing stronger attenuation of high-frequency noise compared to a first-order filter, resulting in a cleaner signal for the comparator and motor control stages. The designed cutoff frequency (~34 Hz) ensured that slow-changing temperature signals were preserved while unwanted noise was effectively removed.

Additionally, this analysis highlighted the impact of real-world non-idealities such as component tolerances, noise, and op-amp limitations, which caused deviations between theoretical, simulated, and experimental results. Despite these differences, the filter performed as expected and contributed to more stable and reliable system behavior.

The analytical cutoff frequency was calculated to be 33.9 Hz. The simulation yielded a cutoff frequency of 33.95 Hz, resulting in a percent error of 0.15%. The experimentally measured cutoff frequency was 34 Hz, corresponding to a percent error of 0.30%. The small deviations are due to component tolerances, non-ideal op-amp behavior, and measurement limitations. Overall, the results show good agreement and confirm the expected second-order low-pass filter behavior.

Complex Power

Description

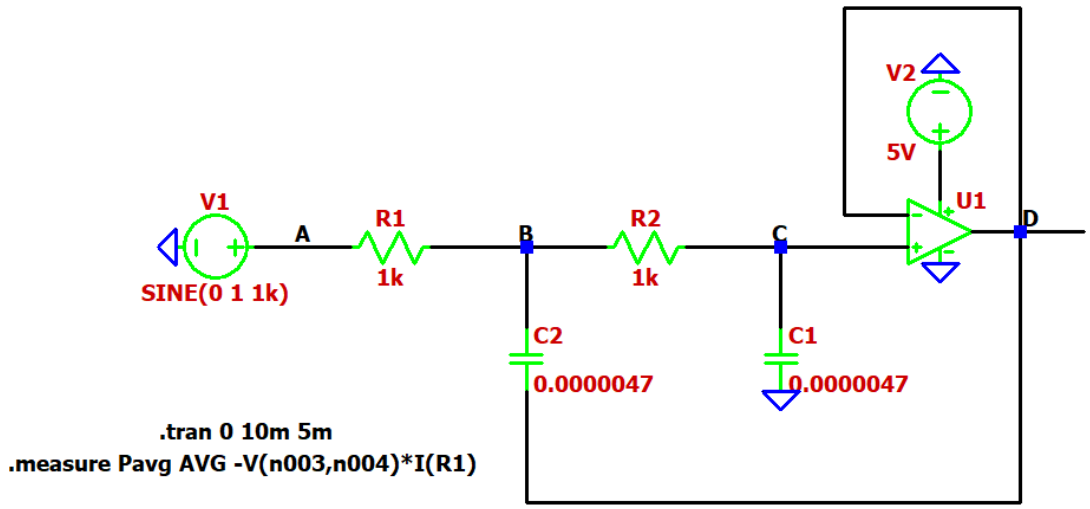
What building block are you applying this concept to?

The concept of complex power is applied to the RC network within the second-order active filter stage. Specifically, the resistors and capacitors in the filter are analyzed as a load driven by an AC source. For simplicity, the filter is approximated as an equivalent RC network at cutoff frequency to analyze power behavior.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The circuit variables derived in this analysis are the complex power (S), real power (P), reactive power (Q), and power factor. Complex power is calculated using the phasor voltage and current, allowing separation of real (dissipated) and reactive (stored) energy in the resistive and capacitive elements.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

Complex Power = S = P + jQ = Vrms*Irms^*
Impedance = Z = R + 1/jωC
I = V/Z
Real Power = P = Vrms*Irms*cos(theta)
Reactive Power = Q = Vrms*Irms*sin(theta)
Pf = cos(theta)
Theta = phase angle between voltage and current = phase(voltage) - phase(current)
Fc = 1/2piRC

Calculate a numerical value for your circuit variable(s).

Given R = 1k, C = 4.7μF
Fc = 1/2pi(1k)(4.7μ) = 34 Hz
Xc = 1 / (2pi f C) = 1 / (2pi * 34 * 4.7u) ≈ 1000 ohms
Z = R - jXc = 1000 - j1000 ohms
|Z| = sqrt(1000^2 + 1000^2) ≈ 1414 ohms
Assume Vrms = 1 V:
Irms = V / |Z| = 1 / 1414 ≈ 0.000707 A
Theta = -45 degrees
P = Vrms * Irms * cos(theta) = 1 * 0.000707 * cos(-45°) ≈ 0.0005 W
Q = Vrms * Irms * sin(theta) = 1 * 0.000707 * sin(-45°) ≈ -0.0005 VAR
S = 0.0005 - j0.0005 VA

|S| = Vrms * Irms = 1 * 0.000707 = 0.000707 VA
Pf = cos(-45°) ≈ 0.707
Since the phase angle is negative, the power factor is leading and capacitive

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

The circuit variables that will be measured are the input voltage (Vin) and the voltage across a known resistor. The current will be calculated using Ohm's Law (I = V/R), and these values will be used to compute real power (P), reactive power (Q), and complex power (S) for comparison with analytical results.

Simulation

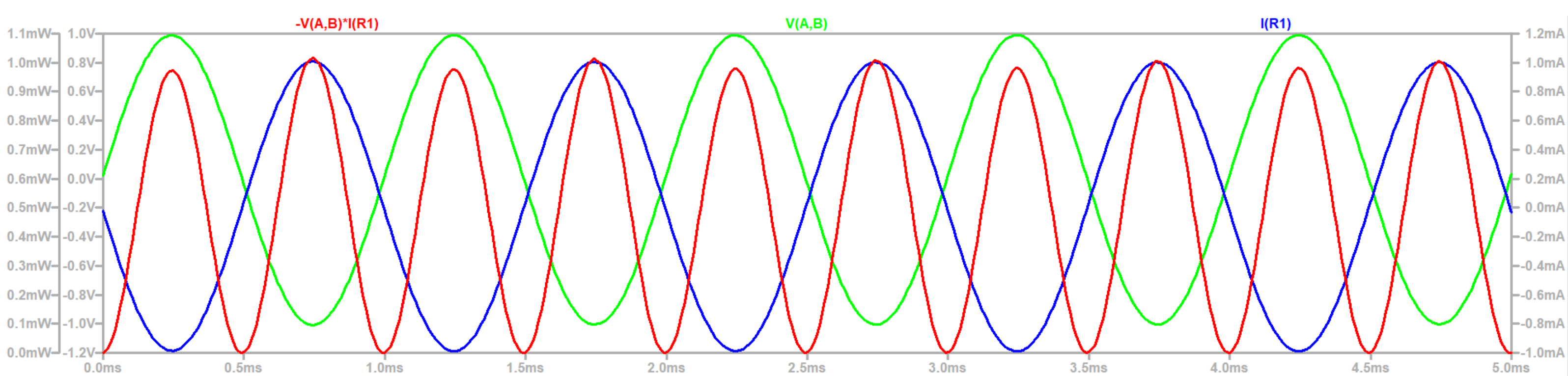
Describe your simulation in 1 to 2 sentences.

A time-domain transient simulation was performed using a sinusoidal input source to observe voltage, current, and instantaneous power in the circuit. The instantaneous power was calculated using waveform math as p(t)=−v(t)i(t)p(t) = -v(t)i(t)p(t)=−v(t)i(t), and the average power was extracted using a .measure command.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

The simulation used a time-domain source defined as SINE(0 1 1k) instead of an AC sweep source. A transient analysis command .tran 0 5ms was added to observe steady-state behavior. Additionally, the power trace was defined as -V(n003,n004)*I(R1) to ensure correct sign convention so that power absorbed by the resistor is positive. The average power was computed using .measure Pavg AVG -V(n003,n004)*I(R1)

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.



SPICE Output Log: C:\Users\Connej3\Desktop\Intro_ECSE\Circuits\Draft4.log

```
LTspice 24.0.12 for Windows
Circuit: * C:\Users\Connej3\Desktop\Intro_ECSE\Circuits\Draft4.asc
Start Time: Wed Apr 29 15:47:49 2026
solver = Normal
Maximum thread count: 16
tnom = 27
temp = 27
method = modified trap
Direct Newton iteration for .op point succeeded.

pavg: AVG(-v(a,b)*i(r1))=0.000496755 FROM 0 TO 0.005

Total elapsed time: 0.090 seconds.
```

Pavg

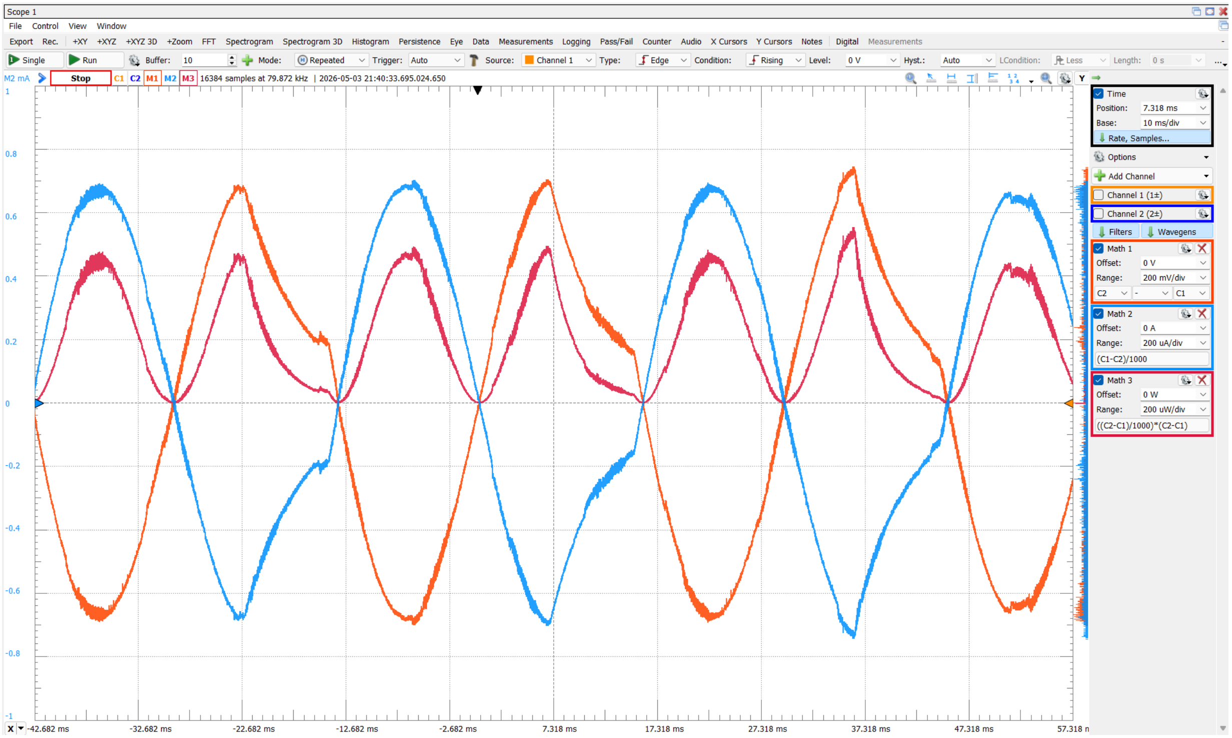
Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

The complex power behavior of the RC filter network was measured experimentally using the Analog Discovery oscilloscope and waveform generator. A 34 Hz sinusoidal input was applied to the circuit, and the voltage across the first resistor was measured to determine current using Ohm's Law. Instantaneous power was then calculated from the measured voltage and current waveforms. The experimental waveforms showed additional noise and slight distortion compared to the analytical and LTspice simulation results due to breadboard parasitics, measurement limitations, and non-ideal op-amp behavior. The circuit utilized an OP484 operational amplifier powered from a ±5 V supply. In order to keep waveform convention consistent between LTspice and our experimental, I flipped the sign convention of the measured current since LTspice makes current values negative.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

In the graph below, the orange trace represents voltage, the blue trace represents current, and the red trace represents power (P=VI).



Discussion

Discuss how this analysis helped you in your design.

This analysis helped verify how energy is distributed within the filter circuit by distinguishing between real power dissipated in resistors and reactive power stored in capacitors. Understanding complex power also provided insight into the phase shift observed in the Bode plot and how the circuit behavior changes with frequency. This was useful in confirming that the filter operates as expected near the cutoff frequency and in understanding the relationship between impedance and frequency response.

The analytical real power was calculated to be 0.0005 W, while the simulation produced 0.000496755 W, resulting in a percent error of 0.649%. The experimentally measured real power was 0.000173 W (P = Vrms*Irms*cos(-45) = (0.495)(0.000495)(0.707)), corresponding to a percent error of 65.4% relative to the analytical value. The discrepancies between analytical, simulation, and experimental results are attributed to non-ideal effects such as component tolerances, parasitic elements, and measurement inaccuracies. Overall, the results show reasonable agreement and validate the expected power behavior of the circuit.

Phasors w/ Wien Bridge Oscillator

Description

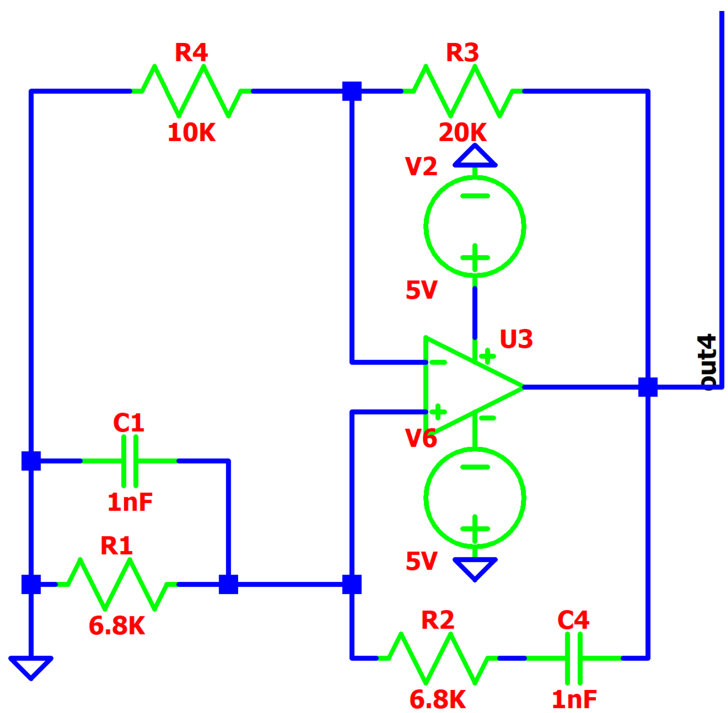
What building block are you applying this concept to?

The Wien Bridge Oscillator Circuit.

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

The output voltage phasor and the transfer function will be derived by analyzing the complex impedances of the circuit components. Using voltage divider equations, we will find the phase shift and frequency to prove the circuit operates as intended.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

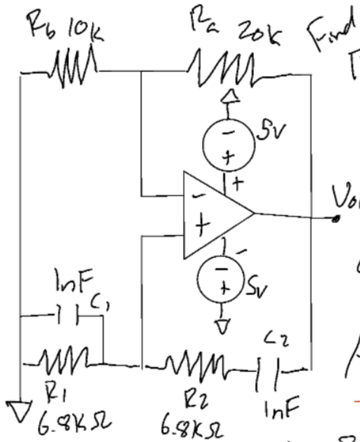
B(s) = V+/Vout

s=jw

w=2*pi*f

Loop Gain(Av) = 1 + R3/R4

Calculate a numerical value for your circuit variable(s).



Find Feedback Transfer function ($\beta(s)$)

$$Z_{series} = R_2 + \frac{1}{sC_2} = \frac{sRC+1}{sC}$$

$$Z_p = \left(\frac{1}{R_1} + sC_1\right)^{-1} = \frac{R}{1+sRC}$$

$$\beta(s) = \frac{V_+}{V_{out}} = \frac{Z_p}{Z_p+Z_s} = \frac{\frac{R}{1+sRC}}{\frac{R}{1+sRC}+\frac{sRC+1}{sC}}$$

$$\beta(s) = \frac{sRC}{s^2R^2C^2+3sRC+1}$$

$$s=j\omega \to \beta(j\omega) = \frac{j\omega RC}{-\omega^2R^2C^2+j\omega 3RC+1} = \frac{1}{3+j\left(\omega RC - \frac{1}{\omega RC}\right)}$$

For a circuit to oscillate, the total phase shift around the feedback loop must be 0°

$$\text{So, } \omega RC - \frac{1}{\omega RC} = 0 \to \omega^2R^2C^2 = 1 \to \omega^2 = \frac{1}{R^2C^2} \to \omega = \frac{1}{RC} = 2\pi f$$

$$f = \frac{1}{2\pi RC} = \frac{1}{2\pi(6.8k)(10^{-9})} = 23405\ Hz \text{ or } 23.4\ kHz$$

$$\beta(j\omega) = \frac{j}{-1^2+j3+1} = \frac{0+j}{0+3j} \to \|Denom\| = \sqrt{0^2+3^2} = 3 \qquad \|Num\| = \sqrt{0^2+1^2} = 1$$

$$\theta_{Denom} = \tan^{-1}\left(\frac{3}{0}\right) = 90° \qquad \theta_{num} = \tan^{-1}\left(\frac{1}{0}\right) = 90°$$

$$\beta(j\omega) = \frac{1\angle 90°}{3\angle 90°} = 0.333\angle(90°-90°) = 0.333\angle 0°$$ attenuation of $\frac{1}{3}$

$$\text{Amplifier gain } (A_v) = 1 + \frac{R_a}{R_b} = 1 + \frac{20k}{10k} = 3\angle 0°$$ zero gain

$$\text{Loop Gain} = A_v + \beta(j\omega) = \frac{1}{3}\angle 0° \times 3\angle 0° = 1\angle 0°$$

Expected Vout will be a sine wave going from 5v to -5v at a frequency of 23.4 kHz centered at 0V

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

We will measure the magnitude of V(out) and the frequency created by this circuit

Simulation

Describe your simulation in 1 to 2 sentences.

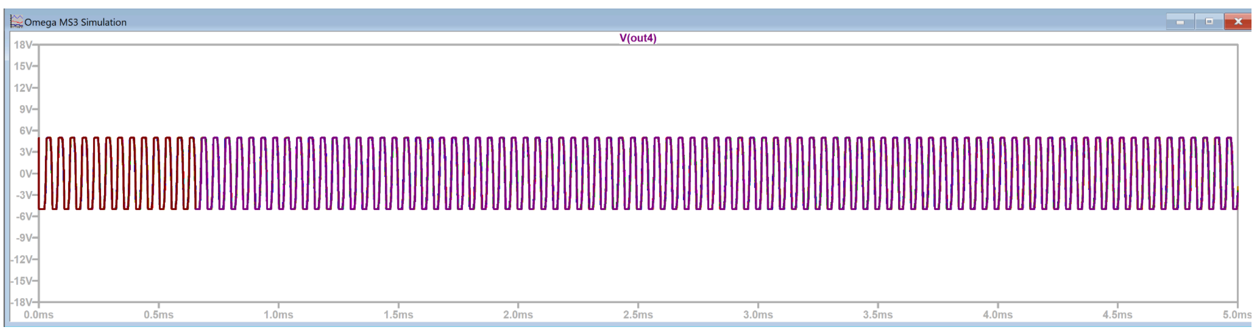
In LtSpice I constructed the Wien Bridge oscillator circuit we used in our project. It consists of 4 resistors with values, 6.8K(2x), 10K, and 20K Ohms, two capacitors with a capacitance of 1nF and LtSpice's Universal Op Amp 2

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

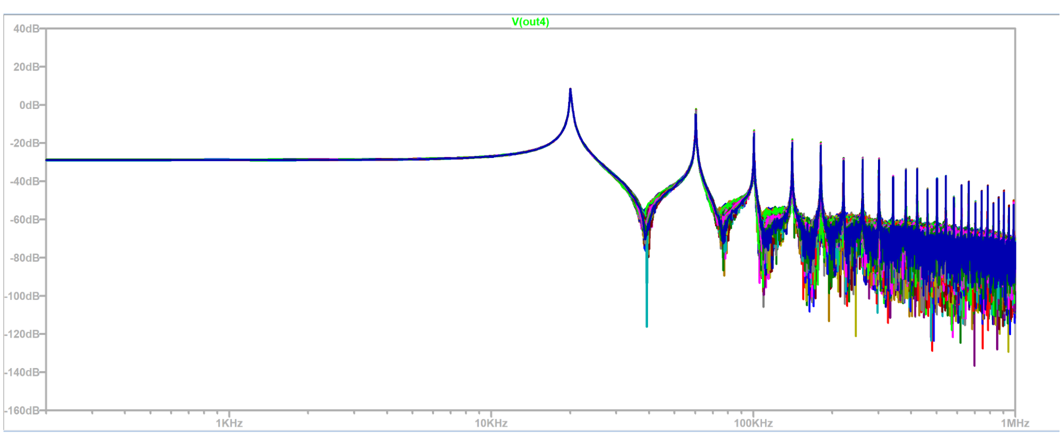
I used a .tran 1m analysis and had to give the simulation a little "kick" by setting V(out4) to 5V. I did this by doing .ic V(out4)=5

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

As you can see the oscillation is exactly what we want. It is fluctuating between 5V and -5V around 0V.



You can also see in the FTT graph, the Highest spike should be the frequency which is just around the expected 23 kHz.



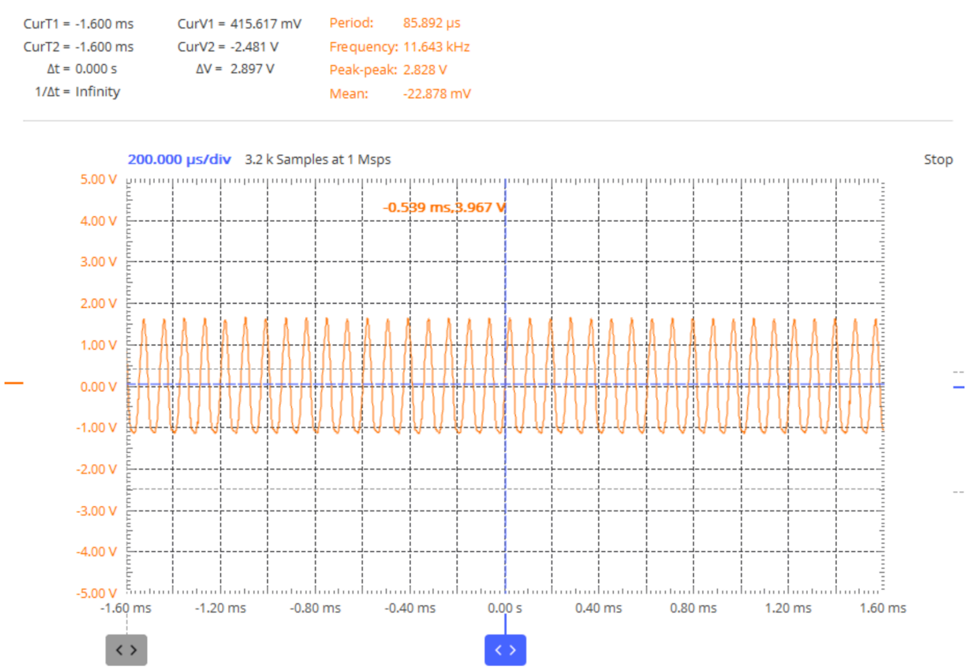
Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

We will measure the output voltage on an oscilloscope to calculate the frequency, mean voltage, and amplitude. The circuit utilizes an OP97 op-amp. The experimental setup isolates Wien Bridge by connecting the positive channel to the output of the oscillator circuit and the negative channel to ground.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

This was the graph and values I got from the experiment. There is a frequency of 11.6kHz, mean of ~0V, and Peak to Peak of 2.8V



Discussion

Discuss how this analysis helped you in your design.

Performing phasor analysis on the Wien Bridge oscillator circuit was useful to understanding the AC Voltage created by the circuit. Without a stable and symmetric sine wave, the motor function will not work optimally or as expected. Although the analysis and simulation support each other, when analyzing the physical circuit, the frequency and peak to peak were a lot less than we expected. Through further research this is because we used an OP97 op amp. The OP97 has a

Slew Rate of 0.2 V/us. This is very low which means that the op amp cannot switch over that quickly with that high of an amplitude. Because of this the peak to peak and amplitude drops. This is very far from the anticipated numbers, but the question is do we care? In this case, the 11.6kHz frequency is fine for controlling the DC motor. The peak to peak also does not matter because we are putting it through a comparator. Because of this, we just need a symmetric oscillating sine wave centered at 0V. This circuit fulfills this requirement, so the circuit still works fine.

The analytical oscillation frequency was determined to be 23.4 kHz. The simulation produced a frequency of 23 kHz, resulting in a percent error of -0.017%. The experimentally measured frequency was 11.6 kHz, corresponding to a percent error of 50.43%. The discrepancy in the experimental result is primarily due to the limited slew rate and non-ideal behavior of the op-amp, which restricts performance at higher frequencies.